

Lecture 1

Course Introduction

Prof. Yinqian Zhang

Spring 2026

General Course Info

- Course Name: **Operating Systems**
- Term: **Spring 2026**
- Time: **Monday 16:20-18:10**
- Location: **107 Lecture Hall I**
- Lecturer: Prof. Yinqian ZHANG (yinqianz@acm.org)
- Lab instructor:
 - Prof. Yinqian ZHANG (yinqianz@acm.org)
 - Ms. Yun SHEN (sheny@mail.sustech.edu.cn)

Self Introduction

- A **professor**
 - of Computer Science and Engineering at SUSTech
 - Previously at Ohio State University in USA
 - Shoot me an email with **subject starting with “[CS302]”** if you have questions
- A **security** researcher
 - Study the security of computer architectures, operating systems, and software applications
 - Find vulnerabilities in CPUs and software
 - Build systems to defend against various attacks
 - **My current research interest:** Build secure systems for confidential computing and AI

Why Are You Here?

- Reason 1: OS is a mandatory course
 - **Yes, you have to pass!**
- *But why do I need to pass?*
 - AI coding is reshaping the industry
 - CS graduates cannot secure jobs
- Well, CS education is **more than just coding**
 - Understanding the theoretical and mathematical foundations that govern what computers can and cannot do.
 - As a CS major, you're not just watching the AI revolution from the sidelines — **you're at the epicenter** of it



Why Are You Here?

- Reason 2: Rumor has it: the course is a breeze to pass
 - **No**, it is **notoriously challenging**, which involves complex concepts like process scheduling, memory management, and synchronization. It requires a strong understanding of C/C++ programming and low-level system interactions.
- *But AI can help with most of the task, right?*
 - **Yes**, but we will make your life **harder**
 - Treat AI as a **collaborator, not** just a **contractor!**



Why Are You Here?

- Reason 3: **Operating system is one of the most important courses for CS majors**
 - **Yes**, it helps you understand how computer systems work
 - understand abstraction and interfaces
 - develop system-level thinking
 - prepare for advanced topics
 - enhance career opportunities



Content of This Course

- We will study the **theory of operating systems**
 - Virtualization, concurrency, persistence
 - Management of CPU, memory, I/O and storage
- We will learn the **implementation of operating systems**
 - Hands-on experience with an educational operating system (xv6)
 - Run your OS on ~~(emulated)~~ RISC-V CPU ~~(via QEMU)~~

Prerequisites

- **C Programming**
 - All course projects will use C (or assembly)
 - **Self-learning is expected!**
- **Data structure and algorithm analysis**
 - It helps you understand OS concepts, e.g., LRU
- **Computer Organization**
 - How do computers work?
 - Hardware and software coordination

Reference Books

- **Operating Systems: Three Easy Pieces**, Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau
 - Virtualization (Process, scheduling, memory address space, swapping)
 - Concurrency (Threads, locks, semaphores)
 - Persistence (I/O, storage, file systems)
 - **Online available: <https://pages.cs.wisc.edu/~remzi/OSTEP/>**
- **Slides**
 - Download from BB before each class
 - **Take notes in class**

Course Structure

- **Lectures** (2 hours per week)
 - Get the main ideas and concepts (mostly in English, may repeat in Chinese)
 - lectures will not follow textbooks (but read textbooks will help)
 - Download slides (before class, and check for updates after)
 - Take notes and ask questions
- **Lab** (2 hours per week)
 - Tutorials on kernel code and lab assignments
 - Do the lab exercises / projects
 - All assignments (even written ones) submitted through labs

Gradings

- Lecture participation: 10%
 - Collect 100 points in quizzes to get full points
 - **No need to request leave in the system.**
- Lab participation: 14%
 - In-class assignments
- Assignment: 16%
 - 5 coding/written assignments
 - Pick best 4 out of 5
- Project: 15% + 5% bonus
 - Teamwork: 2 - 3 students per group
- Mid-term exam: 20%
 - Apr 25, Saturday, 14:00 - 16:00
- Final exam: 25%
 - Closed book (for all exams)
 - No cheating paper (for all exams)

Academic Misconduct

- If an undergraduate assignment is found to be plagiarized, **the first time** the score of **the assignment will be 0.**
- **The second time** the score of **the course will be 0.**
- If a student does not sign the Assignment Declaration Form or cheats in the course, including regular assignments, midterms, final exams, etc., in addition to the grade penalty, the student will not be allowed to enroll in the two CS majors through 1+3, and **cannot receive any recommendation for postgraduate admission exam exemption** and all other academic awards.

Guidelines for AI Uses

- Use AI as a **learning accelerator**, not a **thinking substitute**.
- Acceptable Uses
 - Debugging assistance: "Why is this function throwing an error?"
 - Concept explanation: "Explain recursion like I'm five."
 - Code review: "What edge cases am I missing here?"
 - Refactoring suggestions: "How could I make this more efficient?"
 - Brainstorming: "What are different approaches to this problem?"
- Unacceptable Uses
 - Copy-pasting solutions without understanding them
 - Generating entire assignments to bypass learning
 - Asking AI to do the thinking before you've attempted it yourself
 - Submitting AI-generated code you couldn't explain to a classmate

Tips: How to Pass

- Come to lectures/lab tutorials
 - Take notes, ask questions, take attendance
 - **Pay attention to quiz and in-class lab assignments**
 - **Exams make a big difference in your final grade**
- Visit Blackboard
 - Announcement
 - Lecture slides (before classes, may update afterwards)
 - Assignments and solutions
- Be ready to code!
 - Finish assignments on time
 - **Spend more time on your project!**

OS Concept: Processes

- A process is a program in execution
 - Program is a passive entity and process is an active entity.
- Process needs resources to accomplish its task
 - CPU, memory, I/O, files
 - Process termination requires reclaim of any reusable resources
- Process executes instructions sequentially, one at a time, until completion
 - Single-threaded process has one program counter specifying location of next instruction to execute
 - Multi-threaded process has one program counter per thread
- Typically, system has many processes, some user, some operating system running concurrently on one or more CPUs
 - Concurrency by multiplexing the CPUs among the processes / threads

OS Concept: Process Management

- Creating and deleting both user and system processes
- Suspending and resuming processes
- Providing mechanisms for process synchronization
- Providing mechanisms for process communication
- Providing mechanisms for deadlock handling

OS Concept: Memory

- DRAM (Dynamic Random Access Memory) is the main memory used for all desktop, laptops, servers, and mobile devices
- CPU only directly interacts with the main memory during execution
 - All data in memory before and after processing
 - All instructions in memory in order to execute
- OS manages the main memory for kernel and processes
 - OS dictates which process can access which memory region

OS Concept: Memory Management

- Memory management determines what is in memory when
 - Optimizing CPU utilization and computer response to users
- Memory management activities
 - Keeping track of which parts of memory are currently being used and by whom
 - Deciding which processes (or parts thereof) and data to move into and out of memory
 - Allocating and deallocating memory space as needed

OS Concept: Storage Management

- OS provides uniform, logical view of information storage
 - Abstracts physical properties to logical storage unit - file
 - Each medium is controlled by device (i.e., disk drive, tape drive)
 - Varying properties include access speed, capacity, data-transfer rate, access method (sequential or random)
- File-System management
 - Files usually organized into directories
 - Access control on most systems to determine who can access what
 - OS activities include
 - Creating and deleting files and directories
 - Primitives to manipulate files and dirs
 - Mapping files onto secondary storage
 - Backup files onto stable (non-volatile) storage media

OS Concept: I/O Subsystem

- One purpose of OS is to hide peculiarities of hardware devices from the user
- I/O subsystem responsible for
 - Memory management of I/O including
 - buffering (storing data temporarily while it is being transferred)
 - caching (storing parts of data in faster storage for performance)
 - General device-driver interface
 - Drivers for specific hardware devices

Announcement

- Mid-term exam
 - **Apr 25 (Saturday, Week 9) 14:00 – 16:00**
 - 107/108 Zhihua Building
 - No makeup exams
- A few lectures skipped on Mondays
 - Apr 6 (Week 7)
 - May 4 (Week 11)

Required Reading Materials

- Three easy pieces, CH2,
<https://pages.cs.wisc.edu/~remzi/OSTEP/intro.pdf>

Thank you!

